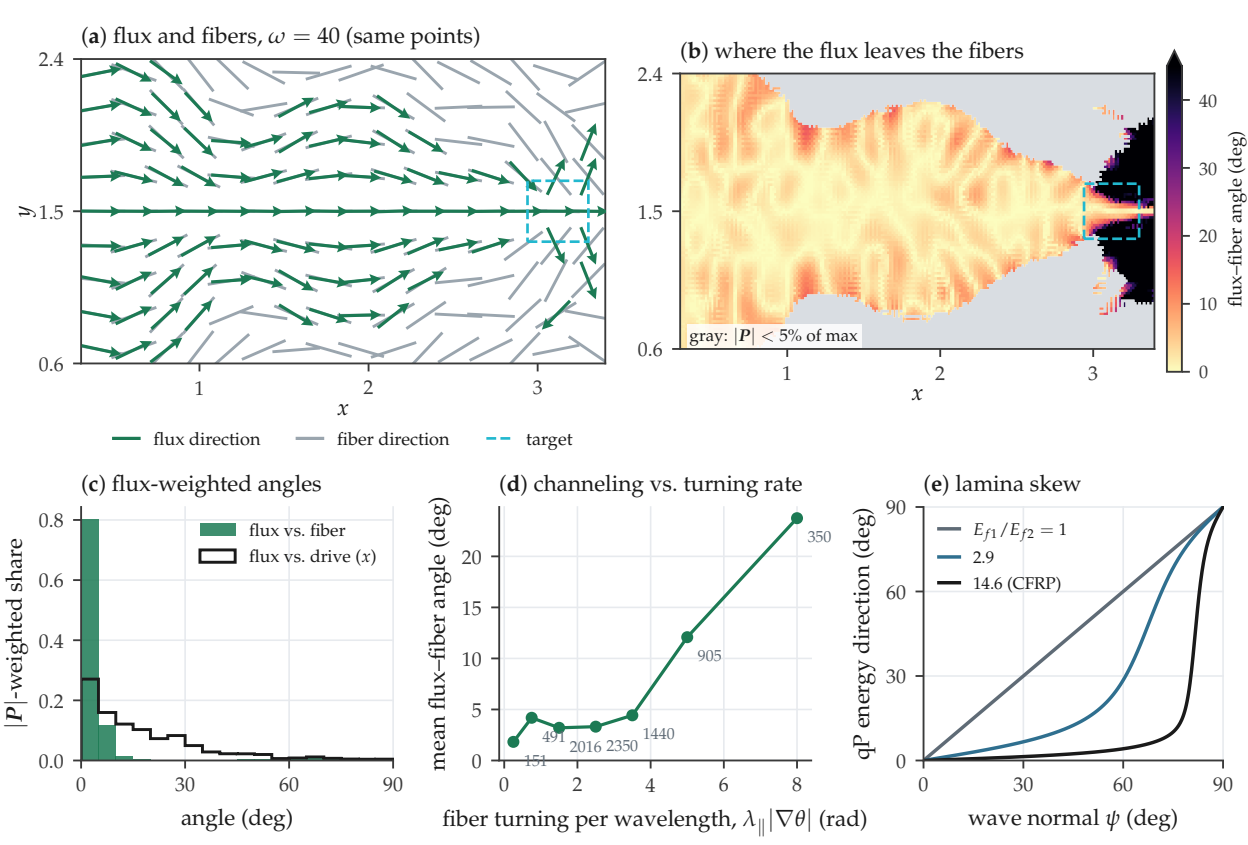

(a) flux and fibers, ω = 40 (same points)
2.4
1.5
0.6
y
1
2
3
x
flux direction
fiber direction
target
(b) where the flux leaves the fibers
gray: |P| < 5% of max
flux–fiber angle (deg)
0
10
20
30
40
(c) flux-weighted angles
flux vs. fiber
flux vs. drive (x)
|P|-weighted share
0.0
0.2
0.4
0.6
0.8
0
30
60
90
angle (deg)
(d) channeling vs. turning rate
mean flux–fiber angle (deg)
0
5
10
15
20
0
2
4
6
8
151
491
2016
2350
1440
905
350
fiber turning per wavelength, λ∥|∇θ| (rad)
(e) lamina skew
qP energy direction (deg)
0
30
60
90
E_f1 / E_f2 = 1
2.9
14.6 (CFRP)
wave normal ψ (deg)